

Experiment 1

Investigating the Significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line Using the Following Parameters:

Resistance (R)
Capacitance (C)
Inductance (L)
Conductance (G)

Test case-1

Code-

```
clear;
clc;
R=10;
L=0.003;
f=2*10^3;
G=4*(10^-6);
C=8*(10^-6);
w=2*pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C1=round(real(Zo));
D1=round(imag(Zo));
printf("Zo = %f + j(%f) ohms\n",C1,D1);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf("Attenuation constant a = %f neper/km\n",a1);
b=imag(P);
b1=round(b*10000)/10000;
printf("Phase constant b = %f radians/km\n",b1);
```

Output: Practical Result

Characteristics Impedance	Attenuation	Phase constant
$20.00 + j(-3)$ ohms	0.2460 Neper/km	1.96350 rad/km

Theoretical Result-

$L = 3 \times 10^{-3} \text{ H}$, $R = 10$
 $f = 2 \times 10^3$
 $G = 0.4 \times 10^{-6}$
 $C = 8 \times 10^{-6}$
 $\omega = 2\pi f = 2 \times 3.1416 \times 2000 = 12566.3 \text{ rad/s}$
 $Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$
 $Z_0 = \sqrt{\frac{(10 + j(12566.3)(3 \times 10^{-3}))}{(0.4 \times 10^{-6} + j(12566.3)(8 \times 10^{-6}))}}$
 $= \sqrt{\frac{10 + j37.70}{0.4 \times 10^{-6} + j0.10053}}$
 $Z_0 = \sqrt{\frac{3.7903 + j0.014}{0.01016 + j0}}$
 $Z_0 = \sqrt{375.06 + j1.44}$
 $= 19.67 + j2.04 \Omega$
 $Z_0 = 19.67 - j2.04 \Omega$
 $\alpha = ?$
 $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$
 $= \sqrt{(10 + j37.70)(0.000004 + j0.10053)}$
 $= -3.79 + j1.0053$
 $\gamma = \sqrt{3.79 + j100.53} = 0.257 + j1.954$
 $\alpha = 0.257 \text{ neper/km}$, $\beta = 1.954 \text{ rad/km}$

Scilab 6.1.1 Console
 $Z_0 = 20.000000 + j(-3.000000)$ ohms
 Attenuation constant $a = 0.256000$ neper/km
 Phase constant $b = 1.963500$ radians/km

```

1 clear;
2 clc;
3 R=10;
4 L=0.003;
5 f=2*10^3;
6 G=0.4*10^-6;
7 C=8*10^-6;
8 w=2*pi*f;
9 Z=R+(j*w*L);
10 Y=G+(j*w*C);
11 Zo=Z/Y;
12 Cl=round(real(Zo));
13 Dl=round(imag(Zo));
14 printf("Zo = %f + j(%f) ohms\n",Cl,Dl);
15 P=Z*Y;
16 a=real(P);
17 al=round(a*10000)/10000;
18 printf("Attenuation constant a = %f neper/km\n",al);
19 b=imag(P);
20 bl=round(b*10000)/10000;
21 printf("Phase constant b = %f radians/km\n",bl);
  
```

Command History
 Command 1: $Z_0 = 20.000000 + j(-3.000000)$
 Command 2: $a = 0.256000$
 Command 3: $b = 1.963500$

Screenshot copied to clipboard
 Automatically saved to screenshots folder.
 Markup and share

Test Case 2:

Code-

```
clear;
clc;
R=10;
L=0.003;
f=2*10^3;
G=4*(10^-6);
C=8*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C1=round(real(Zo));
D1=round(imag(Zo));
printf("Zo = %f + j(%f) ohms\n",C1,D1);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf("Attenuation constant a = %f neper/km\n",a1);
b=imag(P);
b1=round(b*10000)/10000;
printf("Phase constant b = %f radians/km\n",b1);
```

Output- Simulation Result- Variable R

Input(R)	Z0	Attenuation	Phase constant
20	20+j(-5) ohms	0.50020 neper/km	2.010 rad/km
30	21+j(-7) ohms	0.72580 neper/km	2.077 rad/km

Theoretical Result-

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Test Case - 2, Variable R

$L = 3 \times 10^{-3}$, $f = 2000$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$, $R = 20$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$Z_0 = \sqrt{\frac{20 + j(12566.3)(3 \times 10^{-3})}{0.4 \times 10^{-6} + j(12566.3)(8 \times 10^{-6})}}$$

$$= \sqrt{\frac{20 + j37.70}{0.000004 + j0.10053}}$$

$$= \sqrt{375.10 - j98.02}$$

$$Z_0 = 19.65 - j2.49 \Omega$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$= \sqrt{(20 + j37.70)(0.000004 + j0.10053)}$$

$$= \sqrt{-3.78997 + j3.075}$$

$$= 0.734 + j2.054$$

$$\alpha = 0.734 \text{ Neper/Km}$$

$$\beta = 2.054 \text{ rad/Km}$$

Scilab 6.1.1 Console

File Edit Control Applications ?

Scilab 6.1.1 Console

$Z_0 = 20.000000 + j(-5.000000) \text{ ohms}$
Attenuation constant $a = 0.500200 \text{ neper/km}$
Phase constant $b = 2.010000 \text{ radians/km}$

-->

Variable Browser

Name	Value	Type	Visibi...	Mem...
C	8e-06	Double	local	216 B
C1	20	Double	local	216 B
D1	-5	Double	local	216 B
G	4e-06	Double	local	216 B
L	0.003	Double	local	216 B
P	0.5 + ...	Double	local	224 B
R	20	Double	local	216 B
V	4e-06...	Double	local	224 B
Z	20 + ...	Double	local	224 B
Zo	20 - 4...	Double	local	212 B

Command History

11/07/2026 10:27:21 -- //
28/07/2026 09:01:54 -- //

News feed

News feed unavailable.

Output- Simulation Result- Variable l

Input(l)	Z0	Attenuation	Phase constant
0.005	26+j(-6) ohms	0.5845 neper/km	2.58rad/km
0.006	28+j(-5) ohms	0.537 neper/km	2.805rad/km

Theoretical Result-

Given Data :

$$R = 10 \, \Omega$$

$$L = 0.005 \, \text{H (changed)}$$

$$f = 2 \times 10^3 \, \text{Hz}$$

$$G = 4 \times 10^{-6} \, \text{S}$$

$$C = 8 \times 10^{-6} \, \text{F}$$

$$\omega = 2\pi f = 2\pi(2 \times 10^3) = 4000\pi \, \text{rad/s}$$

1) Characteristic Impedance (Z_0) :

$$Z = R + j\omega L = 10 + j(4000\pi)(0.005) = 10 + j62.832 \, \Omega$$

$$Y = G + j\omega C = 4 \times 10^{-6} + j(4000\pi)(8 \times 10^{-6}) = 4 \times 10^{-6} + j0.100531 \, \text{S}$$

$$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{10 + j62.832}{4 \times 10^{-6} + j0.100531}} = 24.11 - j5.68 \, \Omega$$

$$\boxed{Z_0 \approx 24 - j6 \, \Omega}$$

2) Propagation Constant (γ) :

$$\gamma = \sqrt{ZY} = \sqrt{(10 + j62.832)(4 \times 10^{-6} + j0.100531)}$$

$$\gamma = 0.537048 + j2.80457$$

Attenuation constant (α) = Real(γ) = 0.537048 neper/km

$$\boxed{\alpha \approx 0.5370 \, \text{neper/km}}$$

Phase constant (β) = Imag(γ) = 2.80457 radians/km

$$\boxed{\beta \approx 2.8046 \, \text{radians/km}}$$

Final Results :		
$Z_0 \approx 24 - j6 \, \Omega$	Attenuation constant (α) $\approx 0.5370 \, \text{neper/km}$	Phase constant (β) $\approx 2.8046 \, \text{radians/km}$

```

1 clear;
2 clc;
3 R=10;
4 L=0.005;
5 f=2*10^3;
6 G=4*(10^-6);
7 C=8*(10^-6);
8 w=2*pi*f;
9 Z=R+(1i*w*L);
10 Y=G+(1i*w*C);
11 Zo=Z/Y;
12 C1=round(real(Zo));
13 D1=round(imag(Zo));
14 printf("Zo = %f + j(%f) ohms\n",C1,D1);
15 P=Z*Y;
16 a=real(P);
17 b1=round(a*10000)/10000;
18 printf("Attenuation constant a = %f neper/km\n",b1);
19 b=imag(P);
20 b1=round(b*10000)/10000;
21 printf("Phase constant b = %f radians/km\n",b1);
22

```

Command Window Output:

```

Zo = 28.000000 + j(-5.000000) ohms
Attenuation constant a = 0.537600 neper/km
Phase constant b = 2.805100 radians/km

```

Test Case 3:

Code-

```

clear;
clc;
R=10;
L=0.003;
f=2*10^3;
G=4*(10^-6);
C=8*(10^-6);
w=2*pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C1=round(real(Zo));
D1=round(imag(Zo));

```

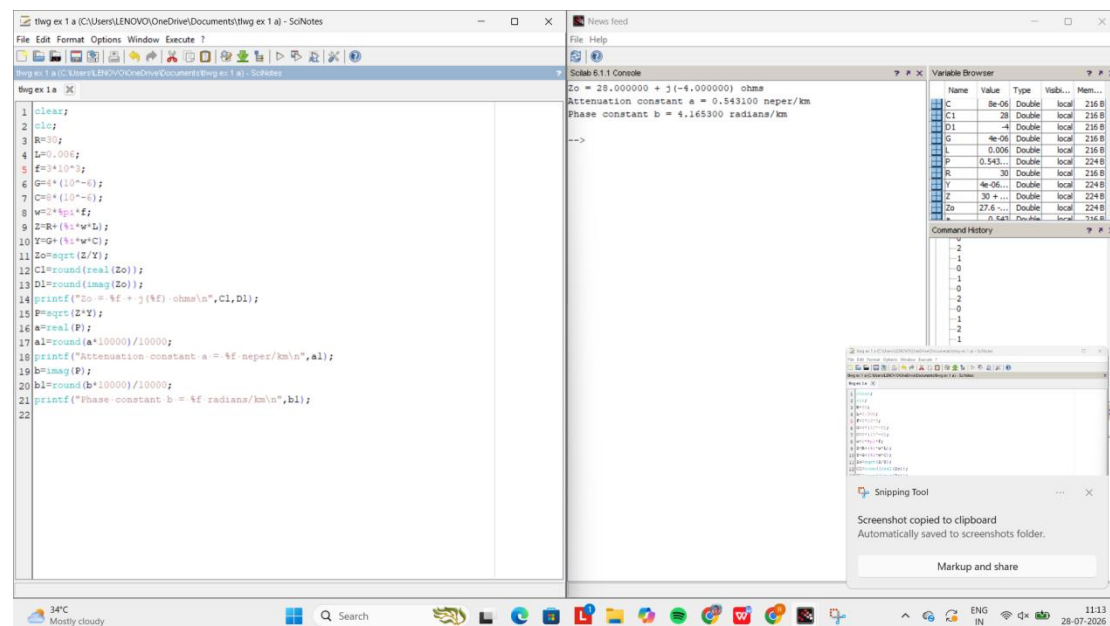
```

printf("Zo = %f + j(%f) ohms\n",C1,D1);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf("Attenuation constant a = %f neper/km\n",a1);
b=imag(P);
b1=round(b*10000)/10000;
printf("Phase constant b = %f radians/km\n",b1);

```

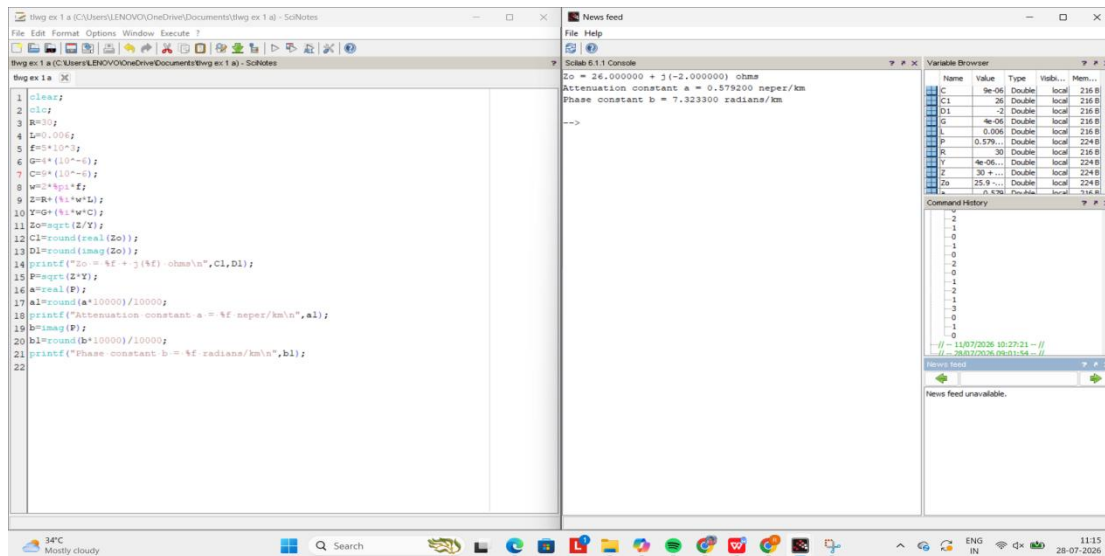
Output- Simulation Result- Variable f

Input(l)	Z0	Attenuation	Phase constant
3kHz	28+j(-4) ohms	0.54310 neper/km	0.543rad/km
5kHz	27+j(-2) ohms	0.546 neper/km	6.90rad/km



Output- Simulation Result- Variable c

Input(c)	Z0	Attenuation	Phase constant
9 micro farad	26+j(-2) ohms	0.579 neper/km	7.323rad/km
11 micro farad	23+j(-2) ohms	0.6403 neper/km	8.096rad/km



Theoretical Result-

Output - Simulation Result -

Variable f

Input (l)	Z_0	Attenuation	Phase constant
3 kHz	$28 + j(-4)$ ohms	0.54310 neper/km	0.543 rad/km
5 kHz	$27 + j(-2)$ ohms	0.546 neper/km	6.90 rad/km

Output - Simulation Result -

Variable c

Input (c)	Z_0	Attenuation	Phase constant
9 micro farad	$26 + j(-2)$ ohms	0.579 neper/km	7.323 rad/km
11 micro farad	$23 + j(-2)$ ohms	0.6403 neper/km	8.096 rad/km